

Acoustic Detection of Impulsive Sounds

[0:00–0:45]

Good morning. // My name is Martin Maxa, from the Department of Measurement here at this faculty.
// I worked on this together with Jakub Svatoš, from the same department. //

Today I will talk about acoustic detection, classification and localization of impulsive sounds, with gunshots as the target class. //

The main focus is gunshot classification: we compare different cepstral features and two classifiers.
// I will also briefly show the localization. //

(// = pause. [CLICK] = advance the slide.)

Acoustic Detection of Impulsive Sounds

└ Motivation

└ Motivation and research questions

Motivation and research questions

- Acoustic detection of gunfire complements camera-based surveillance: the acoustic channel is omnidirectional and does not require line of sight.
- Three required functions, in outdoor environments that are noisy and reverberant:


detection
of the impulsive event


classification
ideally down to calibre


localization
of the source, in 2D

This study

1. Which cepstral representation separates gunshots from acoustically similar false alarms most reliably?
2. Neural network or support vector machine on those features?
3. How does the analysis frame length interact with environmental reflections?

[0:45–1:50] (1/2)

This work is part of the development of an acoustic sensor system for the protection of soft targets.
// Acoustic detection can complement camera-based surveillance. // Sound is omnidirectional, and it does not need a direct line of sight. //

[CUT IF LATE] The complete system should detect an impulsive event, classify it, ideally down to the calibre, and localize its source. // It should also work outdoors, where there is noise and many reflections. **[/CUT]** //

Acoustic Detection of Impulsive Sounds

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3. How does the analysis frame length interact with environmental reflections?

[0:45–1:50] (2/2)

[CLICK]

We focus on three questions. //

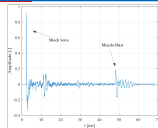
First, which cepstral representation works best for separating gunshots from similar impulsive sounds? // Second, which classifier performs better: a neural network or a support vector machine? // And third, how does the frame length affect the results? //

Acoustic Detection of Impulsive Sounds

└ Signal

└ The acoustic signature of a gunshot

The acoustic signature of a gunshot



One or two components, depending on projectile speed:

- **Shock wave** ("N-wave") - supersonic only ($v > c$); generated along the trajectory, so it arrives first.
- **Muzzle blast** - always present; propellant gases at the barrel, 1-3 ms; here ~45 ms behind.
- Between and after: **reflections**.

[1:50-3:00] (1/2)

Let us first look at the signal of a gunshot. //

A gunshot has one or two components, depending on the projectile speed. // A subsonic shot has only the muzzle blast; a supersonic projectile also produces a shock wave. // This recording is a supersonic shot, so we see both. //

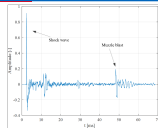
The first component, at about two and a half milliseconds, is the shock wave. // [CUT IF LATE] It is produced along the projectile trajectory and has a characteristic N-shaped pressure profile. [/CUT] //

Acoustic Detection of Impulsive Sounds

└ Signal

└ The acoustic signature of a gunshot

The acoustic signature of a gunshot



One or two components, depending on projectile speed:

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- Between and after: **reflections**.

[1:50-3:00] (2/2)

[CLICK]

The second one, about forty-five milliseconds later, is the muzzle blast. // It comes from the gases at the barrel and is present in every gunshot. //

Between and after these two components, we mainly see reflections from the surroundings. // They matter for the choice of frame length, and I will return to them in the results. //

Acoustic Detection of Impulsive Sounds

└ Signal

└ Single-domain analysis is insufficient

- **Spectral analysis alone** often captures environmental noise rather than the gunshot itself.
- **Time-domain patterns** get distorted by nonlinear dispersion and obstacles.

Consequence

Effective feature extraction must integrate both time and frequency domain processing.

[3:00–3:30]

This also shows why one signal domain alone is not enough. //

The spectrum often describes the environment more than the gunshot itself. // The time-domain shape changes during propagation and because of obstacles. //

So we need features that combine both domains, and cepstral features do exactly that. //

Acoustic Detection of Impulsive Sounds

└ System

└ Proposed system architecture

Proposed system architecture



- Each sensor continuously monitors its surroundings and transmits detected signals to the server for advanced processing.
- Multiple units give broader coverage and adaptability to environments.
- Tested with various firearm calibers and ammunition.

[3:30–4:20]

Now let us look at the proposed system. // It consists of several independent sensor units connected to a remote server. //

Each sensor continuously monitors its surroundings. // When it detects an impulsive event, it sends the recorded signal to the server. // The server then performs two main tasks: classification of the event and localization of the sound source. //

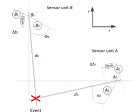
Several units give wider coverage and allow us to estimate the source position. //

The system was tested with several firearm calibres and ammunition types. //

Acoustic Detection of Impulsive Sounds

└ System

└ Localization – bearing from TDoA



$$\theta = \arcsin\left(\frac{c \Delta t}{d}\right)$$

- Two MEMS microphones per unit ($d = 0.186\text{ m}$, $f_s = 44.1\text{ kHz}$) give one bearing θ .
- Two units, two bearings – they intersect at the event.
- Measured: cross-correlation with parabolic interpolation, **MAE 0.77°** over 70 events (-45° to $+60^\circ$).

[4:20–5:20] (1/2)

First, I will briefly explain the localization. //

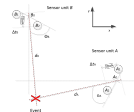
Each sensor unit has two MEMS microphones, about twenty centimetres apart, sampled at 44.1 kilohertz. //

We measure the time difference of arrival between the two microphones. // From this delay, we can calculate the direction to the source using the equation shown here. //

Acoustic Detection of Impulsive Sounds

System

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[4:20–5:20] (2/2)

[CLICK]

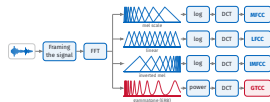
One sensor unit therefore gives us one direction. // With two sensor units, we get two directions, and their intersection gives us the estimated source position. //

We tested this on seventy events at nine reference angles. // Using cross-correlation with parabolic interpolation, we achieved a mean absolute error of zero point seven seven degrees. //

Acoustic Detection of Impulsive Sounds

- Features

- Feature extraction – cepstral coefficients

**[5:20–5:50]**

Now I will move to the main part of the work: classification. //

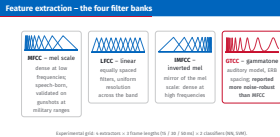
We compared four different cepstral feature extractors. // All of them use the same chain: framing, FFT, a filter bank, and then a logarithm and a discrete cosine transform. //

The main difference is the filter bank. // You can see all four filter banks here, drawn on the same frequency axis. //

Acoustic Detection of Impulsive Sounds

Features

Feature extraction – the four filter banks



[5:50–6:25]

MFCC uses the mel scale, with dense filters at low frequencies. // It comes from speech processing, but it also works for gunshots. //

LFCC uses linearly spaced filters, while IMFCC uses an inverted mel scale, with dense filters at high frequencies. //

GTCC uses a gammatone filter bank based on a model of the human auditory system. // It is also reported to be more robust to noise than MFCC. //

Acoustic Detection of Impulsive Sounds

Results

Classification results – 9 mm class

Classification results – 9 mm class

Frame	Extractor	ACC (%)		BLC (%)		MCC (-)	
		SVM	NN	SVM	NN	SVM	NN
15ms	MFCC	83.90	85.59	80.65	82.76	0.62	0.62
	LFCF	83.90	83.90	80.65	84.52	0.62	0.57
	MFCC	85.59	90.68	83.87	87.92	0.66	0.77
	GTCC	93.22	97.66	87.98	93.95	0.82	0.90
30ms	MFCC	86.75	87.29	82.76	87.92	0.60	0.70
	LFCF	87.29	83.90	78.97	78.97	0.66	0.59
	MFCC	85.59	92.37	82.76	89.65	0.62	0.80
	GTCC	97.66	98.21	93.22	93.95	0.92	0.96
50ms	MFCC	86.75	92.32	76.19	90.32	0.61	0.83
	LFCF	88.08	86.44	77.42	77.42	0.71	0.66
	MFCC	88.83	90.68	76.19	89.65	0.73	0.76
	GTCC	97.66	98.21	93.22	94.77	0.92	0.96

Classification results for the 9mm class. Test partition: 178 events, 31 of them 9mm.

[6:25–6:40]

This table shows the complete classification results for the nine-millimetre class. // The GTCC rows are highlighted. //

I will not go through the numbers here – let us look at the same data in a chart. //

Acoustic Detection of Impulsive Sounds

Results

Classification results – MCC at a glance

**[6:40–7:35] (1/2)**

This chart shows the same results, but only the Matthews correlation coefficient. // MCC considers the whole confusion matrix, so it is less affected by class imbalance than accuracy. //

The red line is GTCC. // It is on top in both panels and for every frame length. //

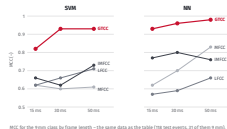
The neural network gives better results than the SVM, and the difference grows with longer frames. //

Acoustic Detection of Impulsive Sounds

Results

Classification results – MCC at a glance

Classification results – MCC at a glance

**[6:40–7:35] (2/2)**

The best configuration is GTCC with the neural network and a fifty-millisecond frame. // It reaches an accuracy of ninety-nine point one five percent, a recall of ninety-six point seven seven percent, and an MCC of zero point nine eight. //

That is one hundred and seventeen correct decisions out of one hundred and eighteen test events. //

Acoustic Detection of Impulsive Sounds

Results

Discussion

- MFCC, LFCC and IMFCC yield similar results.
- GTCC consistently performs better across every frame length and both classifiers.
- NN outperforms SVM - better at modelling complex patterns, especially with longer frames.
- Longer frames improve accuracy, but start to capture environmental reflections, which can pull focus away from the gunshot itself.

Best configuration in the table

GTCC + NN at 50 ms - 99.15 % ACC, 96.77 % RLC, MCC 0.98.

[7:35-8:45] (1/2)

From these results, we can make three main observations. //

First, MFCC, LFCC and IMFCC give similar results, so the filter spacing makes little difference. //

Second, GTCC performs clearly better. // [CUT IF LATE] One possible reason is that the gammatone filter bank is based on an auditory model and is known to be more robust to noise. // However, we did not study this mechanism directly. [/CUT] //

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[7:35-8:45] (2/2)

Third, the neural network performs slightly better than the SVM, especially with longer frames. // This is a general trend, not true for every single result. //

[CLICK]

Finally, longer frames generally improve the classification accuracy. // But with a fifty-millisecond frame, we already start to include more reflections from the environment, not only the gunshot itself. // So even longer frames would not necessarily help. //

Acoustic Detection of Impulsive Sounds

└ Conclusion

└ Conclusion

Conclusion

- A distributed network of stand-alone acoustic sensors with a remote server detects, classifies and localizes impulsive sounds in 20.
- Cepstral coefficients are an effective feature extractor for gunshot classification with both NN and SVM.
- GTCC beats Mel-frequency based coefficients; NN achieves slightly better classification success, especially with longer frames.
- Validated with various firearm calibers and ammunition.

Thank you – questions?

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[8:45–9:30]

To conclude. //

We presented a distributed acoustic sensor system for the detection, classification and localization of impulsive sounds. //

For gunshot classification, cepstral features work well with both tested classifiers. // Among the tested features, GTCC gives the best results. // The neural network also performs slightly better than the SVM, especially with longer frames. //

The system was evaluated on real firearms with several calibres and ammunition types. //

Thank you. //